



Decision Analytics (DAN) User Guide

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1. About this Guide

This user guide describes the Decision Analytics (DAN) application and how to work with it.

2. Overview: Unlocking the Power of Decision Intelligence with Decision Analytics (DAN)

In today's data-driven world, transparency, traceability, and continuous improvement are essential to building trust in automated decision-making. The **Decision Analytics (DAN)** module in *Sapiens Decision* transforms your decision execution data into a powerful asset—enabling deep insights, auditability, and optimization across your decision logic.

DAN acts as a data warehouse for your decision execution history, capturing every input, output, and the exact logic path (Rule Family row ID) that led to each result. This comprehensive data foundation empowers business and technical users alike to explore, analyze, and improve decisions with confidence.

2.1 Key Benefits of DAN

- **Full Transparency & Traceability**

Every decision is stored with its complete context—inputs, outputs, performance metrics, and the logic that determined the outcome—enabling robust audit trails and regulatory compliance.

- **Accelerated Debugging & Root Cause Analysis**

Quickly identify and export individual transactions back to Decision Manager for detailed debugging or build regression test beds for “what-if” analysis, A/B testing, and Champion/Challenger comparisons.

- **Continuous Improvement Through Data**

Use historical decision data to uncover trends, anomalies, and optimization opportunities—driving smarter, faster, and more accurate decision-making over time.

- **Seamless BI Integration**

Connect DAN's structured data views (transactional and snowflake schemas) to your preferred BI tools like **Power BI** or **Tableau** to build custom dashboards and visualizations tailored to your business needs.

2.2 Built-In Dashboards for Instant Insight

DAN includes a suite of intuitive dashboards designed to make your decision data actionable:

- **DAN-AI:** Ask natural language questions about your decision data and get instant answers.
- **Decision Conclusions:** Visualize trends in decision outcomes over time, with support for both textual and numeric result types.
- **Transactions:** Search and filter decisions, then export individual or grouped transactions for further analysis or testing in Decision Manager or other tools.
- **Executions:** Monitor execution statistics across Decision Flows, Decisions, and Input Fact Types.

- **Performance:** Drill into performance metrics to identify bottlenecks or inefficiencies in your decision logic.

2.3 Industry Value: Why Decision Data Matters

In the broader context of enterprise intelligence, DAN delivers the core benefits of a modern **data warehouse and BI environment**:

- **Centralized, High-Quality Data:** A single source of truth for decision execution data, structured for analysis and reuse.
- **Self-Service Analytics:** Empower business users to explore data without relying on IT or data science teams.
- **Faster, Smarter Decisions:** Leverage historical insights to refine logic, reduce risk, and improve outcomes.
- **Scalable Governance:** Ensure decisions are explainable, repeatable, and aligned with business policies and regulations.

3. Setting Up Decision Analytics

To publish data to DAN, you must configure a Kafka subscriber in DE. In addition, set the following required properties:

- **Environment Name:** Defines the environment from which data is published.
- **Tenant ID:** Identifies the tenant associated with the published data.

For configuration instructions, see the **Configuring DE with Kafka subscriber** section in the *Decision Execution (DE) Admin Guide*.

For a description of the message structure published to DAN, see the **Appendix D: Publish and Subscribe Message Format** section in the *Decision Execution (DE) Admin Guide*.

Install and Configure DAN

DAN must be installed and configured to receive published messages from Kafka. This setup enables DAN to process and display Decision data through the analytics dashboard.

For detailed steps, refer to the *Decision Analytics (DAN) Installation and Configuration Guide*.

4. Logging into Decision Analytics

4.1 With Internal Authentication Configuration

You can also log in to DAN using the internal authentication method. If you are using the internal authentication method, you will be redirected to the Decision Manager login page.

To log in to DAN, perform the following:

1. Open your internet browser and enter the URL (`https://{server_url}/dan/login`) in the address bar.
2. In the DAN login window, enter your username and password details and click **login** to navigate to the DAN home page.



DAN Login page

For more information on configuring internal authentication, refer to the **Internal Authentication** section in the *Decision Analytics (DAN) Installation and Configuration Guide*.

5. Decision Analytics Permissions

When you install DAN, there are default permission groups and roles. You can access DAN based on the permissions provided either in development or production mode.

Note: DAN 1.1 is compatible with DM 7.8.21, DM 7.10.x (from 7.10.4 onwards), and DM 7.11.

- For DM 7.8.21 and DM 7.10.x, all permissions except TESTING_WRITE_API are configured in DAN as a file-based configuration. The TESTING_WRITE_API permission is configured in DM.
- From DM 7.11 onwards, all permissions can also be configured directly in DM.

The following permissions define access to key DAN features and dashboards:

Role and Permission Group

- **Role:** DAN Administrator
- **Permission Group:** DAN_Admin

Permissions

- **DAN_AI_READ:** Grants permission to access **DAN-AI**.
- **DAN_DECISION_CONCLUSION_DASHBOARD_READ:** Grants permission to access the **Decision Conclusions Dashboard**.
- **DAN_TRANSACTION_DASHBOARD_READ:** Grants permission to access the **Transactions Dashboard**.
- **DAN_EXECUTION_DASHBOARD_READ:** Grants permission to access the **Executions Dashboard**.
- **DAN_PERFORMANCE_DASHBOARD_READ:** Grants permission to access the **Performance Dashboard**.
- **DAN_SEND_TRANSACTION_TO_DM:** Grants permission to send transaction(s) from **DAN** to **DM**.
- **DAN_DOWNLOAD_TRANSACTION:** Grants permission to download transaction(s).
- **TESTING_WRITE_API:** Grants permission to create test group and test cases by DAN.

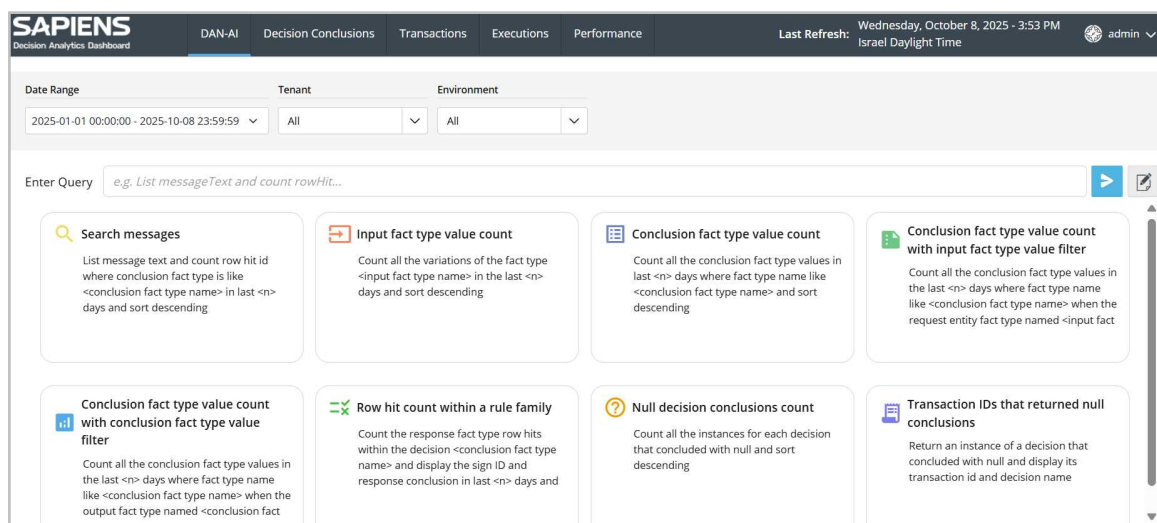
Note: Refer to the **Basic Authentication** and **Internal Authentication** sections in the *Decision Analytics (DAN) Installation and Configuration Guide* to learn how to set up file-based permissions.

6. Decision Analytics Dashboards

Sapiens Decision Analytics (DAN) comes with built-in dashboards to view and analyze data. Each dashboard offers a distinct view of DAN data to meet various use case requirements. Each DAN dashboard is explained below.

6.1 Common Dashboard Features: Headers and Filters

The *Decision Analytics (DAN)* user interface provides consistent and intuitive experience across all dashboards, enabling users to explore decision execution data from multiple perspectives. Two key interface elements—the **Header** and the **Filter Bar**—are shared across all dashboards to support seamless navigation and precise data analysis.



DAN home page

Header

Located at the top of the DAN interface, the **Header** provides essential navigation and context for users:

- **Dashboard Navigation**
Easily switch between dashboards (e.g., DAN-AI, Decision Conclusions, Transactions, Executions, Performance) using the navigation menu in the header. This allows users to explore different analytical views without losing context.
- **Last Refresh Date**
Displays the **date and time of the last data refresh**, based on the **server time zone**.
 - This timestamp helps users determine the freshness of the data they are viewing.
 - It is especially useful when capturing screenshots or exporting data, as it provides a clear reference point for when the data was last updated.

Filter Bar

The **Filter Bar** appears below the header and allows users to refine the data displayed in each dashboard. While some filters may be hidden depending on the dashboard's context, the following options are commonly available:

Standard Filters

- **Date Range**
Define the time window for the data you want to analyze by selecting a start and end date.
- **Tenant**
Filter data by tenant. This is typically a single value for most clients but can be used to segregate data across different Decision Execution (DE) clusters.
- **Environment**
View data from specific environments (e.g., production, test, or different DE clusters and Java Adapters). This helps isolate execution data by deployment context.
- **Asset Type**
Choose the type of asset to analyze:
 - **Decision Flow**
 - **Decision**
 - **Decisions in a Flow** (available on select dashboards)
- **Asset Name**
Filter by the name of a specific Decision Flow or Decision.
- **Decision Name** (*optional*)
For dashboards that support it, this filter allows narrowing the data to a specific Decision within a Decision Flow.
- **Outcome** (*optional*)
Filter transactions by outcome status, such as **Successful** or **Failed**.
- **Metrics** (*optional*)
Select the type of performance metric to view, such as:
 - Full execution time (including data marshalling)
 - Logic execution time only

Advanced Filter Options

Advanced Filter

×

Clear Parameters

Transaction Id

Transaction Id(s)

Input Fact Type

Fact Type Name

Fact Type Value

+

Output Fact Type

Fact Type Name

Fact Type Value

+

Execution Key Value

Key

Value

+

Cancel

Filter

Advanced Filter Options

The **Advanced Filter** expands the filtering capabilities to support more granular and targeted analysis:

- **Transaction ID**
Locate and analyze a specific transaction by its unique identifier.
- **Input Fact Type**
Filter transactions based on specific input values. Multiple input filters can be combined to analyze complex scenarios.
- **Output Fact Type**
Filter based on specific output (response) values. Multiple filters can be applied for detailed result analysis.
- **Execution Key Value**
Filter using custom metadata provided in the execution request.
 - These are user-defined fields (e.g., "State": "NY") that are not part of the decision logic but are included for analytical purposes.
 - This enables powerful segmentation, such as analyzing decision outcomes by region, channel, or customer segment.

Benefits of Common Dashboard Features

- **Consistency:** A unified interface across dashboards ensures a smooth and intuitive user experience.
- **Precision:** Advanced filtering allows users to isolate exactly the data they need for analysis, debugging, or reporting.
- **Context Awareness:** The Last Refresh Date ensures users always know the currency of the data they are viewing.

- **Flexibility:** Filters adapt to the context of each dashboard, showing only relevant options to avoid clutter and confusion.

6.2 DAN-AI Dashboard: Natural Language Insights into Decision Execution Data

Once logged into DAN, you will first be presented with the DAN-AI dashboard.

The **DAN-AI** dashboard in *Sapiens Decision Analytics (DAN)* introduces a powerful and intuitive way to explore your decision execution data—using natural language. With DAN-AI, users can ask questions in plain English and receive immediate, data-driven answers from the DAN data warehouse.

Whether you're investigating anomalies, validating outcomes, or exploring trends, DAN-AI eliminates the need for complex query languages or technical expertise. It bridges the gap between business users and data, making decision intelligence more accessible and actionable than ever.

Key Features

- **Natural Language Query Interface**
Ask questions like “Show all transactions where the conclusion was ‘Declined’ in the last 30 days” or “List the top 5 most common messages by rule hit count.” DAN-AI interprets your intent and returns relevant results instantly.
- **Preset Query Examples**
To help users get started, DAN-AI includes a collection of sample queries such as: *“List message text and count row hit ID where the conclusion fact type is like <conclusion fact type name> in last <n> days and sort descending.”* on the DAN-AI page.
These examples serve as templates for common use cases and can be easily customized.
- **Interactive Exploration**
Results are presented in a clear, tabular format allowing users to refine their insights on the fly.

Benefits of Natural Language Querying

- **Empowers Business Users**
No need to learn SQL or rely on technical teams—anyone can explore decision data using everyday language.
- **Accelerates Analysis**
Quickly surface patterns, outliers, or specific transactions without navigating complex dashboards or reports.
- **Improves Decision Quality**
By making it easier to investigate how and why decisions were made, DAN-AI supports better governance, compliance, and continuous improvement.
- **Boosts Collaboration**

Natural language queries make it easier for cross-functional teams to communicate findings and align on insights.

Query Interface

- **Enter Query:** You can manually enter custom queries in this field. An example of the expected syntax is List messageText and count rowHit.
- **Execute Query Button:** Click the  icon to run the entered query.
- **Open Query Editor:** Click the  icon to enter a new DAN-AI query.

Predefined Query Templates

Below the query input field are predefined query tiles that simplify access to common analytics functions:

- **Search messages:** Retrieves messages and counts based on conclusion fact types.
- **Input fact type value count:** Lists variations of a specific input fact type over a selected period.
- **Conclusion fact type value count:** Counts conclusion fact type values over a defined number of days.
- **Conclusion fact type value count with input fact type value filter:** Filters conclusion fact type counts using a specific input fact type value.
- **Conclusion fact type value count with rule filter**(*partially visible*).
- **Row hit count within a rule family:** Displays the number of times a rule family was hit.
- **Null decision conclusions count:** Counts transactions with null conclusions.
- **Transaction IDs that returned null conclusions:** Retrieves IDs for transactions with null outputs.

You can use predefined query templates to simplify data retrieval without writing complex queries manually. Each tile represents a commonly used query that can be customized as needed. In the example shown below, the Search messages query template has been selected. This template retrieves message text and counts of row hit IDs based on a specified conclusion fact type and date range. The blue-highlighted area in the image indicates the input section where you can manually enter your specific data. Update placeholders such as <conclusion fact type name> and <n> with actual values to tailor the query to your needs. Once you've updated the query input field, select the Execute Query button (blue arrow icon) to generate results based on the entered parameters.

SAPIENS
Decision Analytics Dashboard

DAN-AI | Decision Conclusions | Transactions | Executions | Performance

Last Refresh: Thursday, August 14, 2025 - 11:07 AM British Summer Time [Logout]

Date Range: 2025-01-01 00:00:00 - 2025-08-14 | Tenant: All | Environment: All

Enter Query: List message text and count row hit id where conclusion fact type is like **<conclusion fact type name>** n last **<n>** days and sort descending

Search messages
List message text and count row hit id where conclusion fact type is like <conclusion fact type name> in last <n>

Input fact type value count
Count all the variations of the fact type <input fact type name> in the last <n> days and sort descending

Conclusion fact type value count
Count all the conclusion fact type values in last <n> days where fact type name like <conclusion fact type name>

Conclusion fact type value count with input fact type value filter
Count all the conclusion fact type values in the last <n> days where fact type name like <conclusion fact type name>

Conclusion fact type value count with

Row hit count within a rule familv

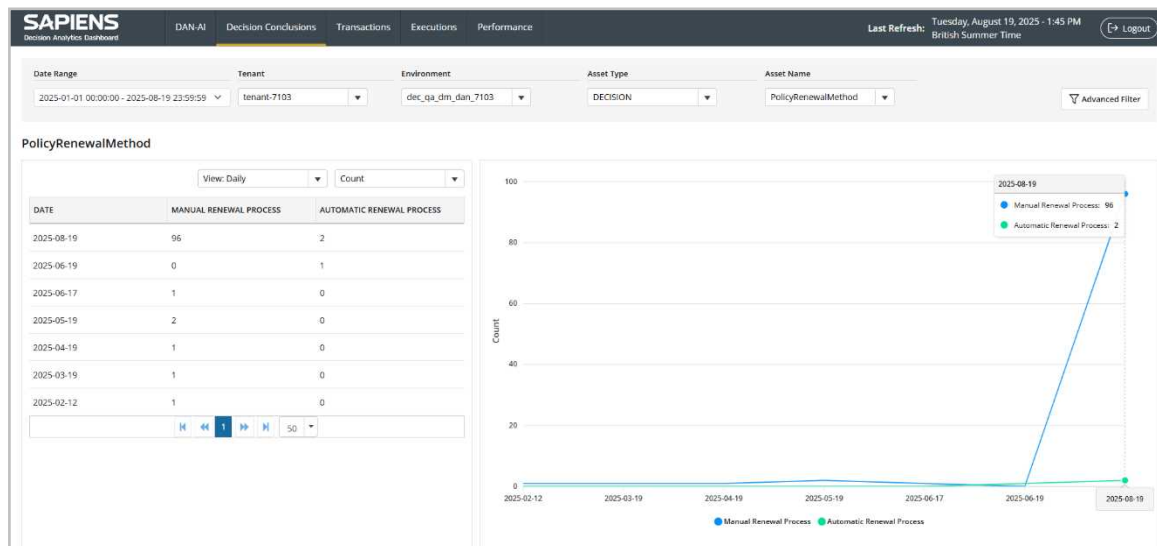
Null decision conclusions count

Transaction IDs that returned null conclusions

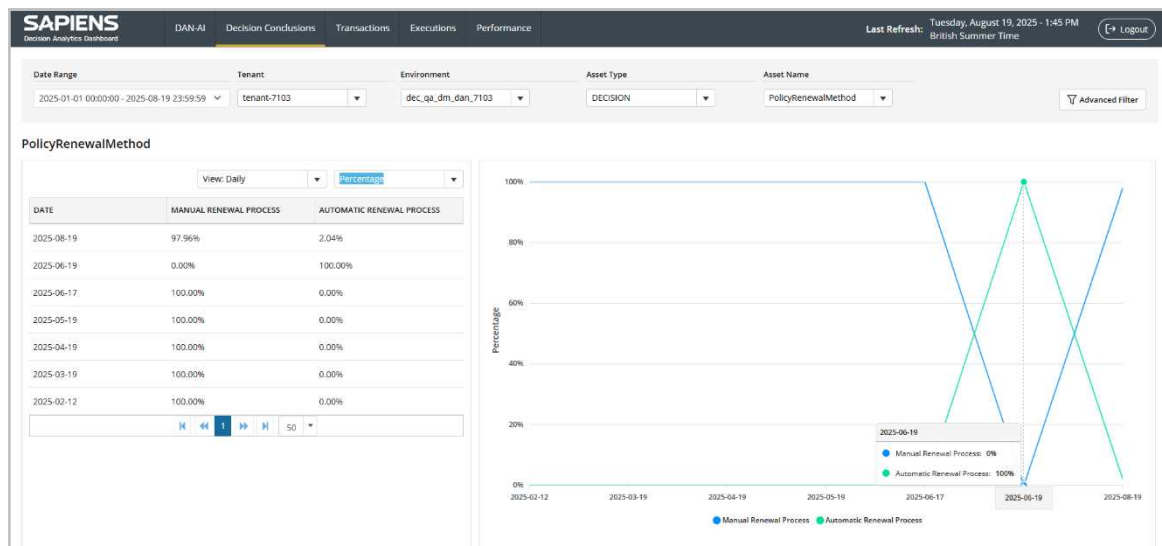
Using predefined query templates

6.3 Decision Conclusions Dashboard: Analyze and Visualize Decision Outcomes over Time

The **Decision Conclusion Dashboard** in *Decision Analytics (DAN)* provides a powerful lens into the outcomes of your deployed Decisions—whether executed independently or as part of a Decision Flow. This dashboard enables users to track, analyze, and visualize how decision results evolve over time, offering valuable insights for auditing, optimization, and business strategy.



Decision Conclusions – Count View



Decision Conclusions – Percentage View

Key Features

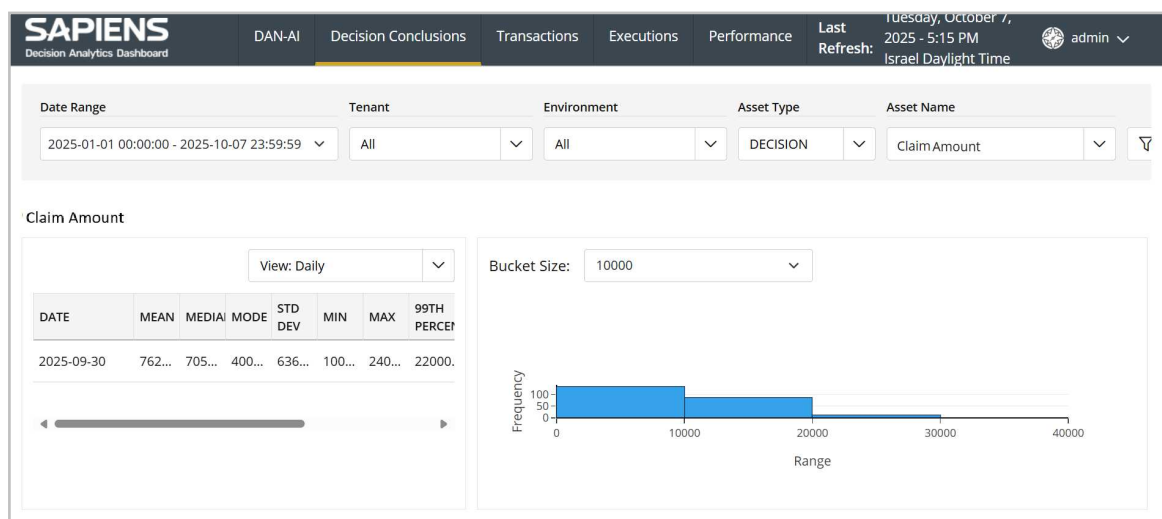
- **Categorical (Text or Code) Results**

For Decisions that return text or coded outcomes, the dashboard displays:

- A **summary table** showing the count of each unique result within the selected time range.
- Time-based breakdowns by **hour, day, week, or month** under **View**.
- The option to view results as **absolute counts** or **percentages** of total transactions.
- A **trend line graph** that visualizes how each result is trending over time, helping users detect shifts in decision behavior.

Numeric Results

For Decisions that return numeric values, the dashboard includes:



Decision Conclusions – Histogram

- A **statistical summary table** for each time period, showing:
 - Mean, median, mode
 - Standard deviation
 - Minimum and maximum values
 - 90th, 95th, and 99th percentiles
- A **histogram** that groups numeric results into value ranges (buckets), with automatic or user-defined bucket sizes (e.g., 10, 100, 1000).

Benefits of Tracking Decision Conclusions

- **Trend Analysis**
Visualize how decision outcomes change over time to identify emerging patterns, seasonal shifts, or unexpected anomalies.
- **Business Insight**
Understand the distribution and frequency of decision results to assess business impact, customer behavior, or policy effectiveness.
- **Data-Driven Optimization**
Use historical result trends to inform logic refinements, improve accuracy, and align decisions with evolving business goals.
- **Audit and Compliance**
Maintain a clear, time-stamped record of decision outcomes for regulatory reporting and internal governance.
- **Performance Monitoring**
For numeric results, statistical summaries help detect outliers or performance drift, supporting continuous improvement.

Some Use Case examples where the Decision Conclusions Dashboard can be used to help improve business operations are as follows:

1. Compliance Monitoring

User: Risk & Compliance Officer

Scenario: A financial institution uses automated decisions to approve or decline loan applications.

Use Case:

The officer uses the dashboard to monitor the frequency of “Declined” decisions over the past quarter. A sudden spike in declines prompts a deeper investigation into whether recent rule changes are unintentionally excluding qualified applicants, helping ensure compliance with fair lending regulations.

2. Business Trend Analysis

User: Product Manager

Scenario: An insurance company uses decision logic to assign risk tiers to new policy applicants.

Use Case:

The manager reviews the trend line of risk tier assignments (e.g., “Low,” “Medium,” “High”) over the past six months. A growing percentage of “High” risk assignments may indicate a shift in applicant profiles or a need to recalibrate risk thresholds.

3. Decision Logic Optimization

User: Decision Modeler

Scenario: A telecom provider uses numeric decision results to calculate customer credit limits.

Use Case:

The modeler uses the histogram and statistical summary to analyze the distribution of credit limits issued. Observing a wide standard deviation and a high 99th percentile, they identify opportunities to tighten logic and reduce outliers, improving consistency and reducing financial exposure.

4. A/B Testing and Champion/Challenger Evaluation

User: Business Analyst

Scenario: A retail company is testing two versions of a discount eligibility decision.

Use Case:

The analyst compares the conclusion trends of both versions over a 30-day period. By viewing the percentage of “Eligible” vs. “Not Eligible” outcomes and their trends, they determine which version aligns better with business goals and customer engagement.

5. Operational Forecasting

User: Operations Manager

Scenario: A logistics company uses decisions to assign delivery priorities.

Use Case:

The manager uses the dashboard to track how many “High Priority” assignments are made daily. A rising trend may indicate increased demand or inefficiencies in routing logic, prompting adjustments in staffing or delivery planning.

6. Customer Behavior Insights

User: Marketing Analyst

Scenario: An e-commerce platform uses decisions to recommend product bundles.

Use Case:

The analyst reviews the frequency of different recommendation outcomes over time. A decline in certain bundle selections may suggest changing customer preferences or the need to refresh product groupings.

6.4 Transactions Dashboard: Explore, Filter, and Use Decision Execution data for testing and debugging

The **Transactions** dashboard in *Decision Analytics (DAN)* provides a comprehensive view of all executed decision transactions across your organization. Displayed in descending order from the most recent to the oldest, this page serves as a dynamic audit trail and a powerful launchpad for testing and debugging decision logic.

With advanced filtering capabilities and seamless integration with *Decision Manager*, the Transactions dashboard empowers users to pinpoint specific decision executions, understand their outcomes, and reuse real-world data to validate and improve decision logic.

SAPIENS
Decision Analytics Dashboard

[DAN-AI](#)
[Decision Conclusions](#)
[Transactions](#)
[Executions](#)
[Performance](#)

Last Refresh: Tuesday, August 19, 2025 - 10:48 AM
British Summer Time

Logout

Date Range: 2025-01-01 00:00:00 - 2025-08-19

Tenant: same_tenant

Environment: dec_qa_pg15_7817

Asset Type: DECISION

Asset Name: Policy Renewal Met

Advanced Filter

Transaction List

Send Group to Decision Manager
Download All

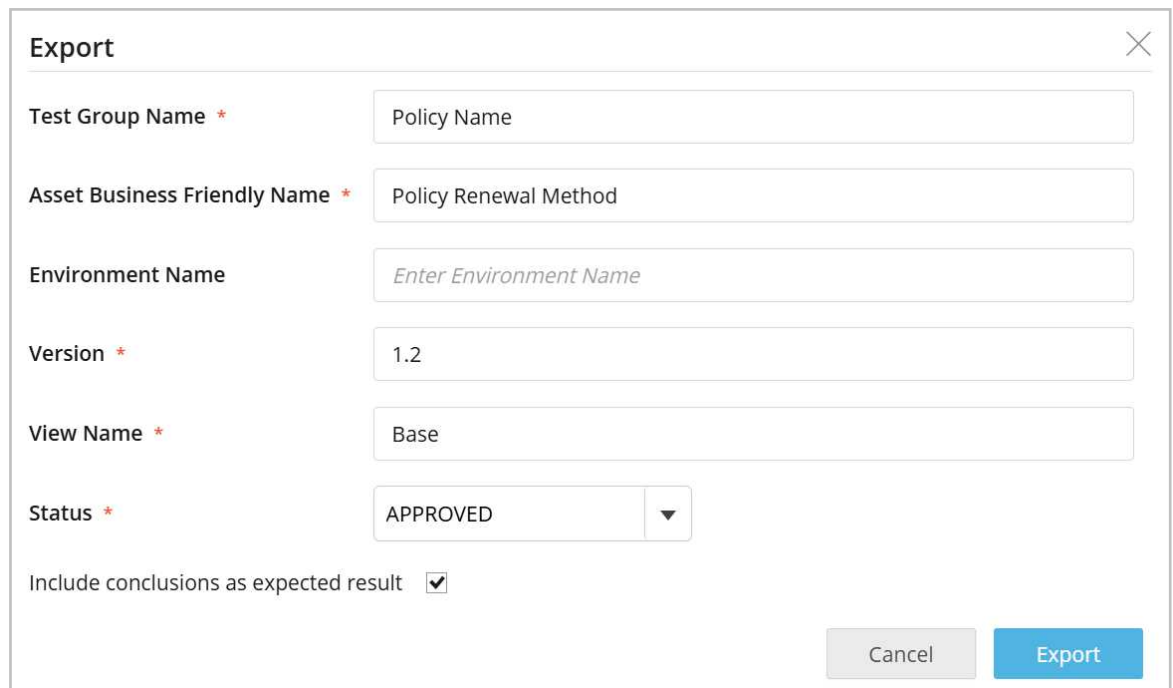
TRANSACTION ID	TRANSACTION DATE	ASSET TYPE	ASSET NAME		
2136125344023756...	2025-08-12 11:59:56	DECISION	Policy Renewal Method	Send to Decision Manager	Download
2136125344275415...	2025-08-12 11:59:56	DECISION	Policy Renewal Method	Send to Decision Manager	Download
2136125341884661...	2025-08-12 11:59:56	DECISION	Policy Renewal Method	Send to Decision Manager	Download
2136125342262149...	2025-08-12 11:59:56	DECISION	Policy Renewal Method	Send to Decision Manager	Download
2136125342597693...	2025-08-12 11:59:56	DECISION	Policy Renewal Method	Send to Decision Manager	Download

1
50

Transactions dashboard

Key Features

- Chronological View of Executions**
 Instantly access the latest decision transactions across all Flows and Decisions, with the ability to drill into each for detailed input, output, and logic traceability.
- Advanced Filtering**
 Use the **Advanced Filter** to narrow down transactions by:
 - Specific Decision Flows or Decisions
 - Input or result values
 - Date ranges or other execution attributes
- Export to Decision Manager**
 - Single Transaction Debugging:** Select any transaction and click **Send to Decision Manager** to send it to *Decision Manager* as a test case using the **Export** dialog. Use the visual debugger to trace the logic path, inspect inputs and outputs, and identify opportunities for improvement.



Export dialog

The **Export** dialog allows users to export test group data associated with a specific decision asset. This dialog includes the following fields that must be completed before initiating the export:

- **Test Group Name (required):** Enter a descriptive name for the test group being exported. For example, Policy Name.
- **Asset Business Friendly Name (required):** The user-friendly name of the asset. This identifies the Decision logic being tested. For example, Policy Renewal Method.
- **Environment Name (optional):** Enter the environment name for the transactions.
- **Version (required):** Enter the version of the asset being exported. For example, 1.2.
- **View Name (required):** Enter the view name for exporting the test group. For example, Base.
- **Status (required):** Specifies the status of the target asset to import the exported data. Select a value from the drop-down list (for example, APPROVED). If you select Draft, the following additional fields appear:
 - Task Name (required): Enter the name of the task.
 - Community Name: Enter the name of the community.
- **Include conclusions as expected result (optional):** Select this checkbox to include Decision conclusions as part of the exported expected results.
- **Cancel:** Closes the dialog without exporting.
- **Export:** Initiates the export process with the specified parameters.

- **Regression Test Groups:** Click **Send Group to Decision Manager** to export a set of filtered transactions as a regression test bed to validate new versions of a Decision or Flow using the **Export Group** dialog. Ideal for **A/B testing, Champion/Challenger testing, or what-if simulations.**

Export Group dialog

Use the **Export Group** dialog to export a set of transactions. This dialog includes the following fields:

- **Test Group Name (required):** Enter a name for the test group. This name will be used to identify the exported set of transactions. For example, Policy Group
- **Asset Business Friendly Name (required):** The readable name of the decision asset being tested. For example, Policy Renewal Method
- **Environment Name (optional):** Enter the environment name for the transactions.
- **Version (required):** Enter the version of the asset being used for export. For example, 1.2
- **View Name (required):** Enter the view name for export. For example, Base
- **Status (required):** Specifies the status of the target asset to import the exported data. Select a value from the drop-down list (for example, APPROVED). If you select Draft, the following additional fields appear:
 - **Task Name (required):** Enter the name of the task.
 - **Community Name:** Enter the name of the community.
- **Export:** Specifies the number of transactions to be exported. The total number of available transactions is shown beside the field. For example, 30 of 30 transactions

- **Include conclusions as expected result (optional):** Select this checkbox to include Decision conclusions as part of the exported expected results.
- **Cancel:** Closes the dialog without performing the export.
- **Export:** Starts the export process using the defined parameters.

Download Transactions

- **Download Icon:** Click the download icon  to export the transaction data individually.
- **Download All:** Click **Download All** to download all visible transactions in the list as a batch export.

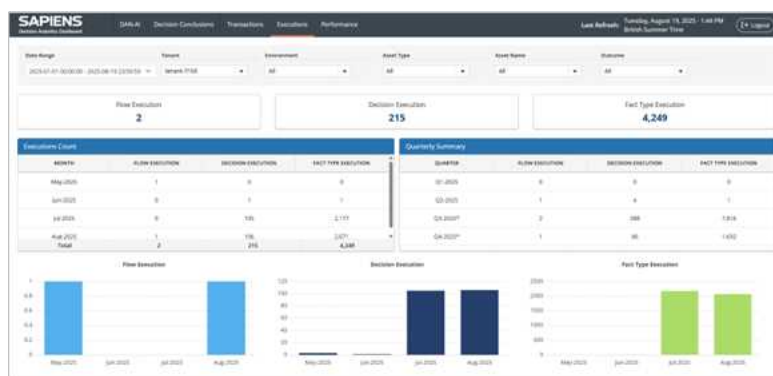
Benefits of Transaction-Based Testing and Debugging

- **Real-World Validation**
Test new logic against actual historical data to ensure accuracy, consistency, and business alignment.
- **Accelerated Debugging**
Quickly isolate and investigate unexpected outcomes using the visual debugger in *Decision Manager*, reducing time-to-resolution.
- **Continuous Improvement**
Leverage past decisions to refine and optimize logic, ensuring your decision models evolve with your business needs.
- **Risk Reduction**
Simulate changes before deployment to avoid regressions and ensure new logic performs as expected under real conditions.

6.5 Executions Dashboard: Monitor and Forecast Decision Activity

The **Executions Dashboard** in *Decision Analytics (DAN)* provides a comprehensive view of how your decision logic is being used over time. It tracks and visualizes the volume of executed **Decision Flows, Decisions,** and **Input Fact Types**, helping users understand usage patterns, audit activity, and forecast future execution trends.

This dashboard is essential for operational transparency, performance monitoring, and licensing management—giving both business and technical users the insights they need to manage decision logic at scale.



Executions Dashboard

Key Features

- **Time-Based Execution History**

View execution counts across customizable time dimensions—**hourly, daily, weekly, or monthly**—to analyze trends and identify peak usage periods.

- **Execution Metrics by Type**

Track execution volumes for:

- **Flows:** End-to-end decision processes.
- **Decisions:** Individual decision components, whether standalone or part of a Flow.
- **Input Fact Types:** Data elements used in decision logic.

- **Visual Trend Analysis**

Interactive **bar charts** display execution trends over time for each category (Flows, Decisions, Fact Types), making it easy to spot spikes, drops, or seasonal patterns.

- **Quarterly Summary & Forecasting**

A dedicated section provides **quarterly execution summaries**, including:

- Historical execution counts
- **Estimated counts** for the current and upcoming quarters.

This helps anticipate future usage and supports proactive planning.

Benefits of the Executions Dashboard

- **Operational Insight**

Understand how your decision logic is being used across the organization, and identify which components are most active.

- **Audit Readiness**

Maintain a clear, time-stamped record of decision executions for compliance and governance purposes.

- **Licensing Forecasting**

Since licensing may be based on executed Decisions and Fact Types, this dashboard helps estimate and manage licensing costs more accurately.

- **Capacity Planning**

Use historical trends and forecasts to plan for future system load, resource allocation, and scaling needs.

6.6 Performance Dashboard: Measure and Monitor Execution Efficiency

The **Performance Dashboard** in *Decision Analytics (DAN)* provides detailed insights into the execution performance of your **Decision Flows** and **Decisions**. By capturing and analyzing execution times, including data marshalling, conversion, and logic processing, this dashboard

helps teams monitor system responsiveness, identify performance bottlenecks, and optimize decision logic for speed and efficiency.

Key Features

- **Comprehensive Performance Metrics**

For each executed Flow or Decision within a selected date range, the dashboard displays:

- **Average execution time**
- **Standard deviation**
- **Minimum and maximum execution times**
- **90th, 95th, and 99th percentile execution times**

All metrics are measured in milliseconds, providing precise visibility into system performance.

- **Interactive Daily Bar Charts**

Visualize performance trends over time with daily bar charts. Hover over any bar to view the full set of performance statistics for that day.

- **Customizable Metric Views**

Easily switch the chart's focus to display different metrics—such as average, standard deviation, or percentile values—by selecting from the options below the chart.

Benefits of Performance Monitoring

- **Operational Optimization**

Identify slow-running decisions or flows and take corrective action to improve system responsiveness and user experience.

- **Proactive Troubleshooting**

Spot anomalies or spikes in execution time early, helping teams investigate and resolve issues before they impact business operations.

- **Capacity and Scalability Planning**

Use historical performance data to forecast system load and plan infrastructure scaling or optimization efforts.

- **Cost Management**

Understand the performance impact of complex logic or high-volume executions, which may influence infrastructure or licensing costs.

- **Business Confidence**

Reliable, fast decision execution builds trust in automated decisioning systems and supports real-time business processes.

7. Decision Analytics Database Schemas

To support robust decision-making, the analytics platform provides two complementary data views: **Transactional View** and **Star/Snowflake Schema View**. Each view serves a distinct purpose, enabling users to explore, analyze, and act on data effectively.

Transactional View

The **Transactional View** presents data in its raw operational format, reflecting how it is captured and executed as a **Flow** or **Decision**. This view is ideal for:

- Detailed auditing and traceability of individual Flows and Decisions.
- Understanding the exact sequence of events.
- Operational reporting and troubleshooting.

It provides high data granularity and fidelity, making it especially useful for pinpointing specific transactions or behaviors.

Star/Snowflake Schema View

The **Star/Snowflake Schema View** is optimized for analytical performance. It structures data into:

- **Fact tables** – quantitative metrics.
- **Dimension tables** – descriptive attributes.

This structure simplifies complex queries, accelerates aggregations, and supports intuitive slicing and dicing across business dimensions such as time, geography, product, or customer. It forms the foundation for dashboards, KPIs, and strategic insights.

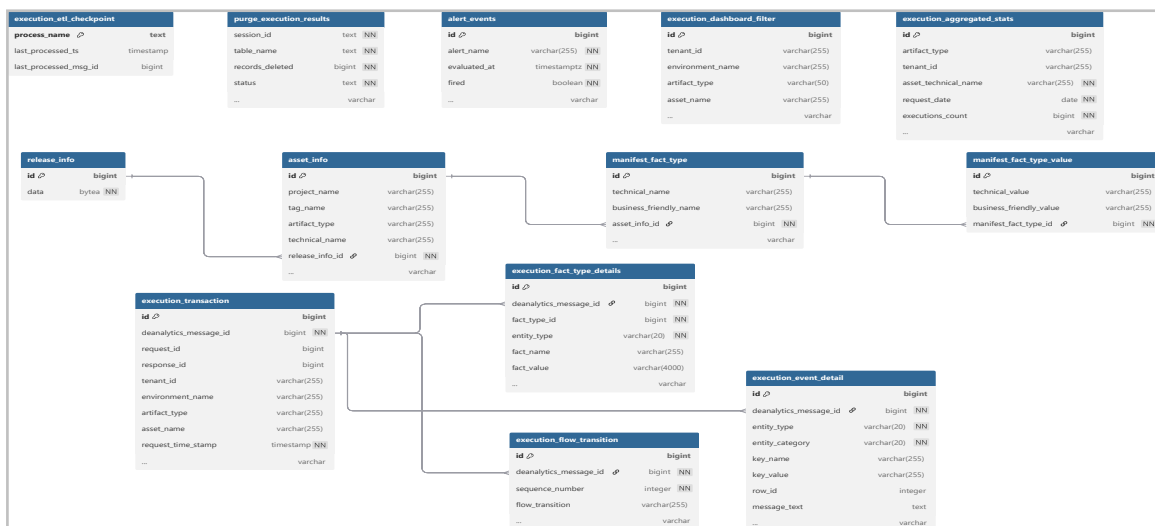
Together, these views empower users to move seamlessly from high-level trends to granular details—enabling both **strategic** and **operational** questions to be answered with confidence. troubleshooting.

7.1. Transactional View Database Schema

DAN 1.5 introduces significant changes to the schema and the mechanism used to persist transactional data. Transactional data is no longer fully normalized at the time of message consumption. In addition, aggregates are no longer processed in a single nightly batch operation.

To enhance application performance and improve the overall user experience, the following design approach is implemented:

- The entity relationships are maintained in the application layer, dropping the cross-table integrity constraints for high throughput.
- For populating UI filter elements such as drop-down lists, the large transaction tables are not queried directly. Instead, dedicated reference tables containing only unique values for the relevant search criteria are populated and refreshed at 15-minute intervals.
- Data aggregation is also performed at 15-minute intervals by default (with the frequency configurable). This enables the nightly aggregation batch process to compute daily aggregates based on these pre-aggregated 15-minute intervals (aggregation of aggregates).



7.1.1 Table Categories

The DAN tables can be broadly categorized into four categories:

1. Execution Analytics Tables
 - a. execution_transaction
 - b. execution_fact_type_details
 - c. execution_event_detail
 - d. execution_flow_transition
 - e. execution_dashboard_filter

- f. execution_aggregated_stats
- g. execution_etl_checkpoint
- 2. Manifest & Metadata Tables
 - a. release_info
 - b. asset_info
 - c. manifest_fact_type
 - d. manifest_fact_type_value
- 3. Alerting Tables
 - a. alert_events
- 4. Administrative Tables
 - a. purge_execution_results

7.1.2 Execution Analytics Tables

These tables store decision execution data for analytics and reporting purposes.

7.1.2.1 execution_transaction

Main transaction table containing decision execution request/response data with denormalized fields for analytics. Each row represents a single decision or flow execution.

Column Name	Data Type	Nullable	Description
id	BIGINT	NO	Auto-generated primary key
deanalytics_message_id	BIGINT	NO	Unique identifier for the analytics message (from the decision engine)
request_id	BIGINT	YES	Reference to the original request
response_id	BIGINT	YES	Reference to the original response
execution_statistics_id	BIGINT	YES	Reference to execution statistics
message_created_at	TIMESTAMP	NO	Timestamp when the analytics message was created
tenant_id	VARCHAR(255)	YES	Tenant identifier for multi-tenant deployments
environment_name	VARCHAR(255)	YES	Environment where the execution occurred (e.g., DEV, QA, PROD)
artifact_type	VARCHAR(255)	YES	Type of artifact executed: DECISION or FLOW
executable_info_id	INTEGER	YES	Reference to the executable information

Column Name	Data Type	Nullable	Description
trace_id	VARCHAR(255)	YES	Distributed tracing identifier
executions_count	INTEGER	YES	Number of executions in this transaction
decisions_instance_count	INTEGER	YES	Count of decision instances executed
flows_instance_count	INTEGER	YES	Count of flow instances executed
bims_instance_count	INTEGER	YES	Count of BIM (Business Information Model) instances
facts_instance_count	INTEGER	YES	Total count of fact instances
input_facts_count	INTEGER	YES	Count of input facts provided
row_hits_count	INTEGER	YES	Count of decision table row hits
row_hits_messages_count	INTEGER	YES	Count of messages generated from row hits
bim_facts_count	INTEGER	YES	Count of BIM-related facts
current_batch_position	INTEGER	YES	Current position in batch execution
total_batch_count	INTEGER	YES	Total count of items in batch
batch_id	VARCHAR(255)	YES	Identifier for batch executions
outcome	VARCHAR(255)	YES	Execution outcome (e.g., SUCCESS, FAILURE)
request_time_stamp	TIMESTAMP	NO	Timestamp when the request was received
service_started	BOOLEAN	YES	Whether the service was started for this execution
cold	BOOLEAN	YES	Whether this was a cold start execution
batch_execution	BOOLEAN	YES	Whether this was part of a batch execution
init_duration	INTEGER	YES	Initialization duration in milliseconds
input_setup_duration	INTEGER	YES	Input setup duration in milliseconds

Column Name	Data Type	Nullable	Description
service_duration	INTEGER	YES	Total service duration in milliseconds
time_to_start_service_duration	INTEGER	YES	Time to start the service in milliseconds
executable_execution_duration	INTEGER	YES	Actual execution duration in milliseconds
executable_construction_duration	INTEGER	YES	Time to construct the executable in milliseconds
result_setup_duration	INTEGER	YES	Result setup duration in milliseconds
request_trace_id	VARCHAR(255)	YES	Request-specific trace identifier
executed_version	VARCHAR(255)	YES	Version of the artifact that was executed
asset_technical_name	VARCHAR(255)	YES	Technical name of the decision/flow asset
asset_business_name	VARCHAR(255)	YES	Business-friendly name of the asset
asset_name	VARCHAR(255)	YES	Computed column: COALESCE(asset_business_name, asset_technical_name)
view	VARCHAR(255)	YES	View name of the asset
version	VARCHAR(255)	YES	Version string of the asset
effective_date	VARCHAR(255)	YES	Effective date used for execution
project_name	VARCHAR(255)	YES	Project name containing the asset
tag_name	VARCHAR(255)	YES	Tag/release name
response_code	INTEGER	YES	HTTP response code of the execution
response_trace_id	VARCHAR(255)	YES	Response-specific trace identifier
request_date	DATE	YES	Date portion of request timestamp
request_hour	INTEGER	YES	Hour of the request (0-23)
request_day_of_week	INTEGER	YES	Day of week (1=Monday, 7=Sunday)
hourly_period	VARCHAR(13)	YES	Hourly aggregation period (format: YYYY-MM-DD HH)

Column Name	Data Type	Nullable	Description
daily_period	VARCHAR(10)	YES	Daily aggregation period (format: YYYY-MM-DD)
weekly_period	VARCHAR(8)	YES	Weekly aggregation period (format: YYYY-WXX)
monthly_period	VARCHAR(7)	YES	Monthly aggregation period (format: YYYY-MM)
last_updated	TIMESTAMP	NO	Timestamp of last record update
de_payload	JSONB	YES	Full decision engine payload as JSON
metadata_processed	BOOLEAN	NO	Flag indicating if metadata enrichment is complete

7.1.2.2 execution_fact_type_details

Stores fact type information extracted from decision executions including input facts, output facts, and decision conclusions.

Column Name	Data Type	Nullable	Description
id	BIGINT	NO	Auto-generated primary key
deanalytics_message_id	BIGINT	NO	Reference to parent transaction
request_id	BIGINT	YES	Reference to the request
response_id	BIGINT	YES	Reference to the response
fact_type_id	BIGINT	NO	Identifier of the fact type
entity_type	VARCHAR(20)	NO	Type of entity: REQUEST or RESPONSE
bim_entity_id	BIGINT	YES	BIM entity identifier
parent_bim_entity_id	BIGINT	YES	Parent BIM entity identifier
is_root_entity	BOOLEAN	YES	Whether this is a root-level entity
bim_entity_name	VARCHAR(255)	YES	Name of the BIM entity
fact_technical_name	VARCHAR(255)	NO	Technical name of the fact
fact_business_name	VARCHAR(255)	YES	Business-friendly name of the fact

Column Name	Data Type	Nullable	Description
fact_name	VARCHAR(255)	YES	Computed column: COALESCE(fact_business_name, fact_technical_name)
fact_technical_value	VARCHAR(4000)	YES	Technical value of the fact
fact_business_value	VARCHAR(4000)	YES	Business-friendly value of the fact
fact_value	VARCHAR(4000)	YES	Computed column: COALESCE(fact_business_value, fact_technical_value)
is_multiple_values	BOOLEAN	YES	Whether the fact has multiple values (list)
is_decision_conclusion	BOOLEAN	YES	Whether this fact is a decision conclusion/output
data_type	VARCHAR(50)	YES	Data type of the fact value
request_time_stamp	TIMESTAMP	NO	Timestamp of the original request
request_date	DATE	YES	Date portion of request timestamp
hourly_period	VARCHAR(13)	YES	Hourly aggregation period
daily_period	VARCHAR(10)	YES	Daily aggregation period
monthly_period	VARCHAR(7)	YES	Monthly aggregation period
metadata_processed	BOOLEAN	NO	Flag indicating if metadata enrichment is complete
last_updated	TIMESTAMP	YES	Timestamp of last record update

7.1.2.3 execution_event_detail

Stores detailed execution information including entity key-values, row hits, and messages from decision executions.

Column Name	Data Type	Nullable	Description
id	BIGINT	NO	Auto-generated primary key
deanalytics_message_id	BIGINT	NO	Reference to parent transaction
request_id	BIGINT	YES	Reference to the request
response_id	BIGINT	YES	Reference to the response
entity_type	VARCHAR(20)	NO	Type of entity: REQUEST or RESPONSE

Column Name	Data Type	Nullable	Description
entity_category	VARCHAR(20)	NO	Category of detail: KEY_VALUE or ROW_HIT
bim_entity_id	BIGINT	YES	BIM entity identifier
parent_bim_entity_id	BIGINT	YES	Parent BIM entity identifier
is_root_entity	BOOLEAN	YES	Whether this is a root-level entity
bim_entity_name	VARCHAR(255)	YES	Name of the BIM entity
key_value_id	BIGINT	YES	Key-value pair identifier (when entity_category = 'KEY_VALUE')
key_name	VARCHAR(255)	YES	Name of the key (when entity_category = 'KEY_VALUE')
key_value	VARCHAR(255)	YES	Value of the key (when entity_category = 'KEY_VALUE')
row_hit_id	BIGINT	YES	Row hit identifier (when entity_category = 'ROW_HIT')
response_fact_type_id	BIGINT	YES	Reference to fact type for row hit
row_id	INTEGER	YES	Decision table row identifier
sign_id	VARCHAR(255)	YES	Signature identifier for the row
row_hit_message_id	BIGINT	YES	Message identifier for row hit
message_text	TEXT	YES	Message text generated by row hit
message_category	VARCHAR(255)	YES	Category of the message
request_time_stamp	TIMESTAMP	NO	Timestamp of the original request
request_date	DATE	YES	Date portion of request timestamp
hourly_period	VARCHAR(13)	YES	Hourly aggregation period
daily_period	VARCHAR(10)	YES	Daily aggregation period
monthly_period	VARCHAR(7)	YES	Monthly aggregation period
last_updated	TIMESTAMP	NO	Timestamp of last record update

7.1.2.4 execution_flow_transition

Captures flow transition sequences showing the execution path through flow decisions.

Column Name	Data Type	Nullable	Description
id	BIGINT	NO	Auto-generated primary key
deanalytics_message_id	BIGINT	NO	Reference to parent transaction
response_id	BIGINT	YES	Reference to the response
flow_transition_id	BIGINT	YES	Flow transition identifier
sequence_number	INTEGER	NO	Order of the transition in the flow execution
flow_transition	VARCHAR(255)	YES	Name/identifier of the transition step
trace_id	VARCHAR(255)	YES	Trace identifier for this transition
request_time_stamp	TIMESTAMP	NO	Timestamp of the original request
request_date	DATE	YES	Date portion of request timestamp
hourly_period	VARCHAR(13)	YES	Hourly aggregation period
daily_period	VARCHAR(10)	YES	Daily aggregation period
monthly_period	VARCHAR(7)	YES	Monthly aggregation period
last_updated	TIMESTAMP	NO	Timestamp of last record update

7.1.2.5 execution_dashboard_filter

Pre-aggregated filter values for dashboard dropdown menus. The filter type is determined by which columns are populated.

Column Name	Data Type	Nullable	Description
id	BIGINT	NO	Auto-generated primary key
tenant_id	VARCHAR(255)	YES	Tenant identifier
environment_name	VARCHAR(255)	YES	Environment name
tag_name	VARCHAR(255)	YES	Tag/release name
project_name	VARCHAR(255)	YES	Project name
artifact_type	VARCHAR(50)	YES	Artifact type: DECISION or FLOW

Column Name	Data Type	Nullable	Description
outcome	VARCHAR(50)	YES	Execution outcome
asset_technical_name	VARCHAR(255)	YES	Technical name of the asset
asset_business_name	VARCHAR(255)	YES	Business-friendly name of the asset
asset_name	VARCHAR(255)	YES	Computed column: COALESCE(asset_business_name, asset_technical_name)
asset_data_type	VARCHAR(50)	YES	Data type of the asset
flow_technical_name	VARCHAR(255)	YES	Technical name of the flow
flow_business_name	VARCHAR(255)	YES	Business-friendly name of the flow
flow_name	VARCHAR(255)	YES	Computed column: COALESCE(flow_business_name, flow_technical_name)
is_decision_in_flow	BOOLEAN	YES	Whether this decision is embedded in a flow
last_updated	TIMESTAMP	NO	Timestamp of last record update

7.1.2.6 execution_aggregated_stats

Daily aggregated execution statistics per tenant, environment, asset, and outcome. Used for high-level dashboard metrics and trend analysis.

Column Name	Data Type	Nullable	Description
id	BIGINT	NO	Auto-generated primary key
artifact_type	VARCHAR(255)	YES	Type of artifact: DECISION or FLOW
tenant_id	VARCHAR(255)	YES	Tenant identifier
outcome	VARCHAR(255)	YES	Execution outcome
batch_execution	BOOLEAN	YES	Whether these are batch executions
environment_name	VARCHAR(255)	YES	Environment name
asset_technical_name	VARCHAR(255)	NO	Technical name of the asset
asset_business_name	VARCHAR(255)	YES	Business-friendly name of the asset

Column Name	Data Type	Nullable	Description
artifact_name	VARCHAR(255)	NO	Name of the artifact
project_name	VARCHAR(255)	YES	Project name
tag_name	VARCHAR(255)	YES	Tag/release name
request_date	DATE	NO	Date for which stats are aggregated
executions_count	BIGINT	NO	Total number of executions
decisions_instance_count	BIGINT	NO	Total decision instances
flows_instance_count	BIGINT	NO	Total flow instances
bims_instance_count	BIGINT	NO	Total BIM instances
facts_instance_count	BIGINT	NO	Total fact instances
input_facts_count	BIGINT	NO	Total input facts
row_hits_count	BIGINT	NO	Total row hits
row_hits_messages_count	BIGINT	NO	Total row hit messages
bim_facts_count	BIGINT	NO	Total BIM facts
rows_aggregated	BIGINT	NO	Number of source rows aggregated
max_request_time_stamp	TIMESTAMP	NO	Latest request timestamp in the aggregation
created_at	TIMESTAMP	NO	When the aggregation record was created
updated_at	TIMESTAMP	NO	When the aggregation record was last updated

Unique Constraints: artifact_type, tenant_id, outcome, batch_execution, environment_name, asset_technical_name, project_name, tag_name, request_date

7.1.2.7 execution_etl_checkpoint

Tracks ETL process state for incremental data processing jobs.

Column Name	Data Type	Nullable	Description
process_name	TEXT	NO	Primary key - name of the ETL process
last_processed_ts	TIMESTAMP	YES	Timestamp of last processed record
last_processed_msg_id	BIGINT	YES	Last processed message ID for incremental processing

7.1.3 Alerting Tables

7.1.3.1 alert_events

Stores alert evaluation events for monitoring and notification purposes.

Column Name	Data Type	Nullable	Description
id	BIGINT	NO	Auto-generated primary key
alert_name	VARCHAR(255)	NO	Name of the alert rule
evaluated_at	TIMESTAMPTZ	NO	Timestamp when the alert was evaluated
fired	BOOLEAN	NO	Whether the alert condition was triggered
computed_value	DOUBLE PRECISION	YES	The computed metric value that was evaluated
message	TEXT	YES	Alert message or description
recipients	TEXT	YES	List of notification recipients

7.1.4 Administrative Tables

7.1.4.1 purge_execution_results

Stores results from data purge operations for audit and monitoring purposes.

Column Name	Data Type	Nullable	Description
session_id	TEXT	NO	Unique identifier for the purge session
table_name	TEXT	NO	Name of the table that was purged
records_deleted	BIGINT	NO	Number of records deleted
execution_time_ms	BIGINT	NO	Time taken for the purge operation in milliseconds

Column Name	Data Type	Nullable	Description
status	TEXT	NO	Status of the purge: SUCCESS or FAILED
error_message	TEXT	YES	Error message if status is FAILED
purge_timestamp	TIMESTAMP	YES	Timestamp when the purge was executed

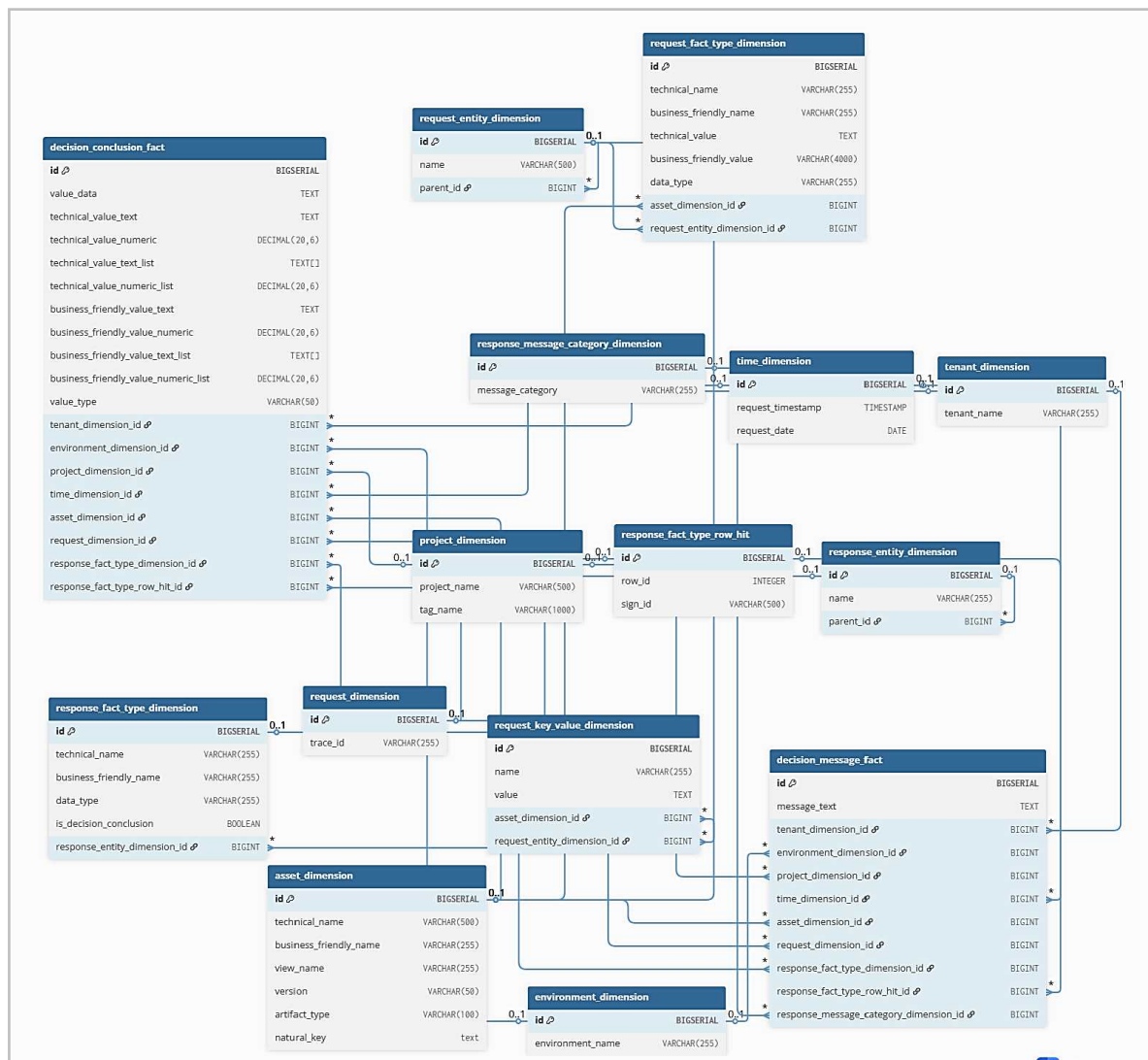
7.1.5 Time Period Columns

Several tables contain pre-computed time period columns for efficient time-based aggregations:

Column Name	Format	Example	Description
hourly_period	YYYY-MM-DD HH	2024-01-15 14	Hour-level grouping
daily_period	YYYY-MM-DD	2024-01-15	Day-level grouping
weekly_period	YYYY-WXX	2024-W03	ISO week-level grouping
monthly_period	YYYY-MM	2024-01	Month-level grouping

7.2. Snowflake View Database Schema

The database schema provided is a highly normalized, analytics-focused design built around a star-schema pattern, combining multiple dimension tables and two main fact tables. Dimension tables such as tenant, environment, project, asset, and various entity and fact-type tables provide rich contextual metadata about requests, responses, and system assets. The central fact tables—`decision_conclusion_fact` and `decision_message_fact`—record detailed decision outcomes and messages, referencing all relevant dimensions to enable flexible and powerful querying. Hierarchical relationships in entity dimensions support complex parent-child modeling, while unique constraints and optimized indexes ensure data integrity and performance. This schema is well-suited for advanced reporting, auditing, and analytics scenarios in multi-tenant, multi-environment systems where traceability and contextual understanding of decisions are essential.



7.2.1 time_dimension

Stores timestamps and dates associated with requests, providing a reference for time-based analysis.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
request_timestamp	TIMESTAMP	Stores Request Time
request_date	DATE	Stores Request Date

7.2.2 tenant_dimension

Represents the tenant (customer, organization, or logical group) for which data is recorded. Each tenant has a unique name.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
tenant_name	VARCHAR(255)	Name of the tenant

7.2.3 environment_dimension

Defines the environment context (such as "production", "staging", or "development") in which events or data occur.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
environment_name	VARCHAR(255)	Name of the environment

7.2.4 project_dimension

Represents a project context, including both the project name and an optional tag for further categorization or versioning.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
project_name	VARCHAR(500)	Name of the Project
tag_name	VARCHAR(1000)	Tag Name

7.2.5 request_dimension

Captures unique request traces, typically used for tracking and correlating requests throughout the system.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
trace_id	VARCHAR(255)	Unique Trace Id of the request

7.2.6 asset_dimension

Describes a technical or business asset within the system, including names, views, versions, and artifact types. A generated key ensures uniqueness across key identifying columns.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
name	VARCHAR(500)	Asset Name
view_name	VARCHAR(255)	Name of the view this asset belongs to
version	VARCHAR(50)	Asset Version
artifact_type	VARCHAR(100)	Artifact Type whether Decision or Flow

7.2.7 request_entity_dimension

Defines entities involved in a request, supporting hierarchical relationships via self-referencing parent IDs (e.g., parent-child entities within a request).

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
name	VARCHAR(500) NOT NULL	Name of the BIM in the request
Parent Id	BIGINT REFERENCES request_ entity_dimension(id)	Self-referencing relationship with parent BIMs
name	VARCHAR(500) NOT NULL	Name of the BIM in the request

7.2.8 request_fact_type_dimension

Describes fact types associated with requests, including technical and business names and values. Links to an asset and optionally to a request entity.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
name	VARCHAR(255) NOT NULL	Name of the request Fact Type
value	TEXT	Value of the Fact Type

Column Name	Data Type	Description
asset_dimension_id	BIGINT REFERENCES asset_dimension(id)	Reference to the asset this fact type belongs to
request_entity_dimension_id	BIGINT REFERENCES request_entity_dimension(id)	Reference to the Request BIM this Fact Type belongs

7.2.9 request_key_value_dimension

Captures key-value pairs relevant to requests, optionally linked to specific assets and entities for more granular analysis.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
name	VARCHAR(255) NOT NULL	Request Key name
value	TEXT	Request Key value
asset_dimension_id	BIGINT REFERENCES asset_dimension(id)	Reference to the asset this key value pair belongs to
request_entity_dimension_id	BIGINT REFERENCES request_entity_dimension(id)	Reference to the Request BIM this key value pair belongs to

7.2.10 response_entity_dimension

Defines entities involved in a response, supporting hierarchical/self-referencing relationships similar to request entities.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
name	VARCHAR(255) NOT NULL	Name of the Response BIM
Parent ID	BIGINT REFERENCES response_entity_dimension(id)	Self-referencing relation with the Parent BIM entity

7.2.11 response_fact_type_dimension

Describes fact types associated with responses, including names, data types, and whether the fact represents a decision conclusion. Linked to a response entity.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
name	VARCHAR(255) NOT NULL	Name of the Response Fact Type

Column Name	Data Type	Description
is_decision_conclusion	BOOLEAN DEFAULT FALSE	Boolean value to indicate if this Fact Type is a conclusion
response_entity_dimension_id	BIGINT REFERENCES response_entity_dimension (id)	Reference to the Response BIM entity this Fact Type belongs to

7.2.12 response_fact_type_row_hit

Represents a "row hit" in response facts, with a row identifier and signature, likely used for detailed traceability of data processing.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
row_id	INTEGER	Numeric Row ID
sign_id	VARCHAR(500)	Logical Sign/Row Hash

7.2.13 response_message_category_dimension

Lists categories for response messages, enabling categorization and grouping of responses or messages by type.

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
message_category	VARCHAR(255) NOT NULL	Message Category

7.2.14 decision_conclusion_fact

Central fact table recording decision outcomes. Contains technical and business values (including text, numeric, and list forms), value types, and links to all relevant dimensions (tenant, environment, project, time, asset, request, response fact type, and row hit).

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
value_data	TEXT	Conclusion Fact data type
value_text	TEXT	Conclusion Fact Text value
value_numeric	DECIMAL(20)	Numeric Value of the conclusion Fact
value_text_list	TEXT[]	List Value of the conclusion Fact

Column Name	Data Type	Description
value_numeric_list	DECIMAL(20)	Numeric List Value of the conclusion Fact
value_type	VARCHAR(50)	Value Type
tenant_dimension_id	BIGINT REFERENCES tenant_dimension(id)	Reference to the Tenant Dimension
environment_dimension_id	BIGINT REFERENCES environment_dimension(id)	Reference to the Environment Dimension
project_dimension_id	BIGINT REFERENCES project_dimension(id)	Reference to the Project Dimension
time_dimension_id	BIGINT REFERENCES time_dimension(id)	Reference to the Time Dimension
asset_dimension_id	BIGINT REFERENCES asset_dimension(id)	Reference to the Asset Dimension
request_dimension_id	BIGINT REFERENCES request_dimension(id)	Reference to the Request Dimension
response_fact_type_dimension_id	BIGINT REFERENCES response_fact_type_dimension(id)	Reference to the Response Fact Types
response_fact_type_row_hit_id	BIGINT REFERENCES response_fact_type_row_hit(id)	Reference to the Row Hit dimension

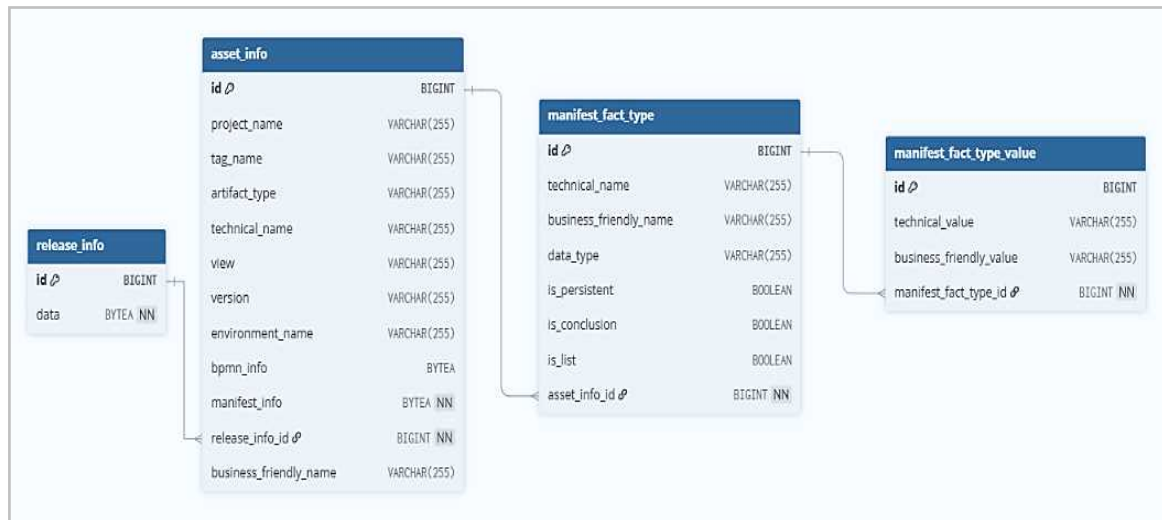
7.2.15 decision_message_fact

Central fact table for messages generated as part of a decision process. Stores the message text and links to all pertinent dimensions (tenant, environment, project, time, asset, request, response fact type, row hit, and message category).

Column Name	Data Type	Description
id	BIGSERIAL PRIMARY KEY	Primary Key
message_text	TEXT	Message text
tenant_dimension_id	BIGINT REFERENCES tenant_dimension(id)	Reference to the tenant dimension
environment_dimension_id	BIGINT REFERENCES environment_dimension(id)	Reference to the environment dimension
project_dimension_id	BIGINT REFERENCES project_dimension(id)	Reference to the project dimension
time_dimension_id	BIGINT REFERENCES time_dimension(id)	Reference to the time dimension

Column Name	Data Type	Description
asset_dimension_id	BIGINT REFERENCES asset_dimension(id)	Reference to the asset dimension
request_dimension_id	BIGINT REFERENCES request_dimension(id)	Asset to the request dimension
response_fact_type_dimension_id	BIGINT REFERENCES response_fact_type_dimension(id)	Reference to the Response Fact Type dimension
response_fact_type_row_hit_id	BIGINT REFERENCES response_fact_type_row_hit(id)	Reference to the Row Hit dimension
response_message_category_dimension_id	BIGINT REFERENCES response_message_category_dimension(id)	Reference to the message category dimension

7.3. Metadata Database Schema [Common for Transactional and Snowflake]



The following tables have been added to support metadata:

7.3.1 release_info

The first table that stores the release information before normalizing to asset and manifest level details.

Column Name	Data Type	Description
id	Int8	Primary Key
data	Bytea	Release information in Blob

7.3.2 asset_info

Column Name	Data Type	Description
id	Int8	Primary Key
project_name	varchar(255)	Name of the project
tag_name	varchar(255)	Tag name
artifact_type	varchar(255)	Decision/Flow
technical_name	varchar(255)	Technical name of the asset conclusion
view	varchar(255)	View

Column Name	Data Type	Description
version	varchar(255)	Version
environment_name	varchar(255)	Name of the execution environment
bpmn_info	Bytea	BPMN info file in blob format
manifest_info	Bytea	Manifest info in blob
release_info_id	Int8	Foreign Key – references the release info table

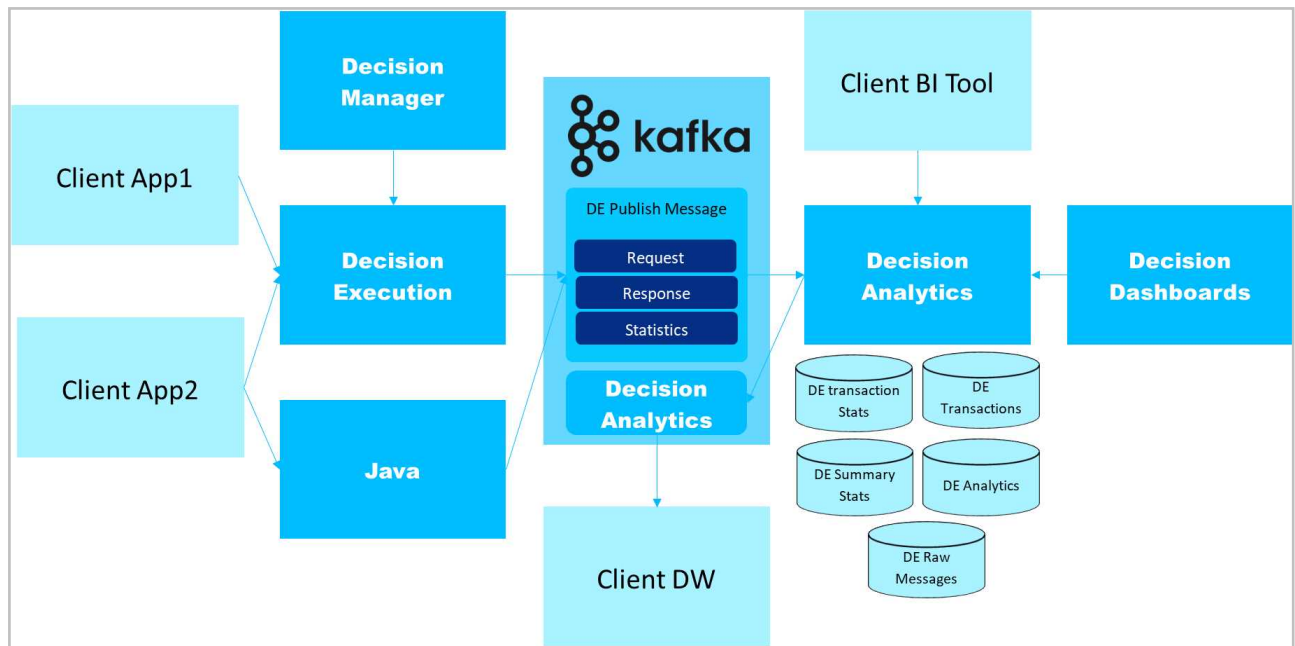
7.3.3 manifest_fact_type

Column Name	Data Type	Description
id	Int8	Primary Key
technical_name	varchar(255)	Technical name of the fact type
business_friendly_name	varchar(255)	Business name
data_type	varchar(255)	Data Type
is_persistent	boolean	TRUE If persistent Fact Type
is_conclusion	Boolean	TRUE if conclusion
is_list	Boolean	TRUE if list Fact Type
asset_info_id	Int8	Foreign key – references the asset_info table

7.3.4 manifest_fact_type_value

Column Name	Data Type	Description
id	Int8	Primary Key
technical_value	varchar(255)	Technical value
business_friendly_value	varchar(255)	Business value
Manifest_fact_type_id	Int8	Foreign key – references the manifest_fact_type table

8. Decision Analytics Architecture



The diagram above illustrates the end-to-end flow of the Decision Analytics (DAN) process.

When a decision is executed—either through Decision Execution (DE) or Java code generated via the Java Adapter—the request, response, and associated statistics can optionally be published to a Kafka topic. This publication is controlled through configuration settings in both DE and the Java Adapter. For detailed configuration instructions, refer to the *Decision Execution (DE) Admin Guide* and the *Decision Manager (DM) Adapters Guide*.

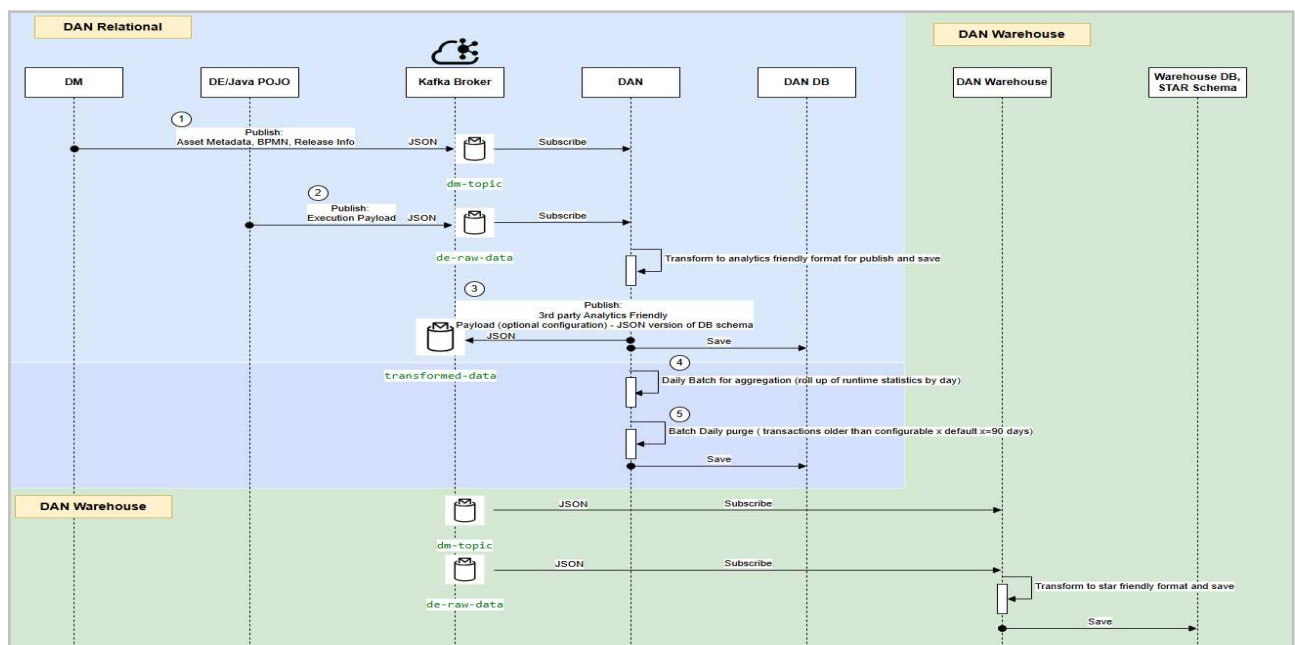
Once messages are published to Kafka, DAN is configured to consume these messages. Upon receipt, DAN transforms them into standardized Decision Analytics messages and republishes them to a dedicated DAN Kafka topic. These messages can then be:

- Consumed by customer analytics platforms for further analysis, or
- Processed by DAN for ingestion into the DAN database.

8.1. Data Views in the DAN Database

DAN persists multiple views of decision execution data to support various analytical and operational use cases:

- **Transaction Statistics:** Captures detailed execution and performance metrics for individual transactions.
- **Transaction Summary Statistics:** Provides daily aggregated statistics to optimize storage and improve query performance.
- **Transactional Data View:** A normalized schema designed for querying detailed transactional data.
- **Star/Snowflake Schema View :** A dimensional model optimized for high-performance analytical queries across multiple dimensions.
- **Raw Request Transaction Messages:** Stores original JSON request payloads, enabling easy extraction for use in Decision Manager as test cases or regression test groups.



Sequence of events

8.1.1 Asset Metadata Publication

- **DM** publishes asset metadata, BPMN (process definitions), and release information.
- This is sent as **JSON** messages to the Kafka **dm-topic** [configurable].

8.1.2 Execution Payload Publication

- The Java or DE execution environment publishes execution payloads (runtime execution data).
- These are sent as JSON messages to the Kafka **de-raw-data** topic [configurable].

Kafka holds messages in topics (**dm-topic**, **de-raw-data**). DAN system subscribes to these topics for further processing.

8.1.3 Publish Transformed Data

- DAN transforms raw execution data into an analytics-friendly format. The transformed payload (still JSON) is published to Kafka **transformed-data** topic.
- Transformed data is saved into the DAN DB.

8.1.4 Daily Aggregation Batch

- Data is aggregated (rolled up) daily by runtime statistics (e.g., daily totals, counts, averages).
- This can be scheduled using a cron expression via a configuration (decision.dan.execution.statistics.scheduler.job.cron).
- Default is every midnight.

8.1.5 Daily Transactional View Schema Purge

- Old transactions (older than a configurable number of days, default = 90) are purged to keep the system lean.
- This can be scheduled using a cron expression via a configuration (decision.dan.purge.transaction.batch.cron)
- Default is 3 AM everyday.

8.2 Synching Decision Manager Meta Data with DAN

Metadata is sent from DM to DAN during release deployments to DE or Java Execution environments via a Kafka messaging system. DAN consumes these messages to keep its metadata up to date. The same is patched for any old DE transactions also that were processed before the asset metadata was received in DAN from DM.

DAN maintains metadata associated with the release deployments and assets, including:

- **Release Deployment Information**
 - Project Name,
 - Tag Name,
 - Environment Name,
 - Release Info
 - List of Assets
- **Asset Metadata [For Each asset in the list]**
 - Artifact Type (Decision or Flow)
 - Asset Name
 - View
 - Version

- BPMN Info
- Manifest Info

The missing metadata for the transactions that were processed before the metadata was received from DM is updated via two mechanisms:

- An on-demand REST API call `[/api/metadata/patch]`
- When metadata is received from DM, we check the existing transactions from DE/Java execution that miss the metadata and update retrospectively. This behavior can be turned off and on via a configuration:
`spring.kafka.dm.trigger.metadata.scheduler.job.cron.enabled`

Note: If there are assets already deployed to DE or Java, it will be necessary to do a redeployment to get the metadata published to DAN for these asset versions.

8.3 Integration with Dashboards and BI Tools

The DAN database powers the built-in DAN Dashboards, which offer a set of predefined visualizations for monitoring and analysis. Additionally, customers can connect their own Business Intelligence (BI) tools to the DAN database to create custom dashboards and perform advanced queries. This enables deeper insights into decision outcomes and helps identify opportunities for optimization.

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